Zohaib Ali Mughal

Full-Stack Software Engineer: TypeScript, Next.js, NestJS, Flutter

z.mughal.24648@gmail.com | Karachi, Pakistan
zohaibmughal.dev | github.com/Zohaib24648 | linkedin.com/in/zohaibmughal24648

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Dart, SQL, Python
Frontend: Next.js, React, Tailwind CSS, TanStack Query, Radix / shadcn UI
Mobile: Flutter (BLoC, clean architecture), Capacitor, App Store and Play Store releases
Backend: Node.js, NestJS, Fastify, tRPC, REST / OpenAPI, Zod, BullMQ, NATS JetStream
Data and DevOps: PostgreSQL, Prisma, Supabase, Redis, Docker, AWS, Railway, Vercel, GitHub Actions
Payments and Quality: Stripe Connect, Paymob, Airwallex, Sentry, Jest, Vitest, Playwright, Testcontainers

PROFESSIONAL EXPERIENCE

Slidey LLC | *Software Engineer II* **Nov 2024 – Present**

ETG Platform | Software Engineer II portal.evolutiongroup.net.au | case study **May 2026 – Present**

Field-service platform for an Australian electrical, security and telecom group (two entities sharing customers), with admin, client, supplier and technician mobile portals. Greenfield build to stop an estimated $5k a week lost to unbilled supplier costs and billing from scheduled rather than actual hours.

- Designed the 25-package Turborepo (Next.js frontend, Fastify API, BullMQ worker): 145 tRPC routers and 1,000+ procedures over a 212-model Prisma schema covering dispatch, timesheets, assets, quoting and billing; all money as BigInt cents.
- Enforcement-first decision engine: one pure, unit-testable gate checks every dispatch, closeout and invoice transition, returns structured failure codes, and logs every block and manager override (8 live checks on the dispatch calendar).
- Scored two-hop reconciliation of bank transactions against supplier invoices; a job cannot be invoiced until its costs are matched.
- Five role-scoped portals over one API across 148 page routes: internal operations, technician mobile, client and subcontractor self-service, and tokenised public links for quotes, invoices and feedback.
- Supplier integration layer: typed adapters for eight external systems, including three trade portals with no public API (SAP Commerce and Magento storefronts) fetched and parsed server-side, plus ABN lookup and two payment feeds.
- Credential security: envelope encryption through Google Cloud KMS, Argon2 hashing, and permission-scoped procedures rather than route-level guards.
- Infrastructure: docker-compose local parity, GitHub Actions CI building GHCR images, Railway and Vercel deploys with post-deploy smoke tests; 30,000+ tests across 2,200+ files.

Stack: Next.js, React, Fastify, tRPC, Prisma, PostgreSQL, BullMQ, Redis, Better Auth, Docker.

Hyss | Software Engineer I hyss.io | case study **Jan 2026 – Apr 2026**

Live-event staffing marketplace: companies hire certified freelance crew, with competency-matched shifts, call sheets, realtime chat, invoicing and payouts. One Next.js codebase serves web, iOS and Android through Capacitor.

- Type-safe Turborepo: 93 Prisma models, about 340 tRPC procedures, plus a native Android Credential Manager plugin for Capacitor.
- Stripe Connect core: payout onboarding, hybrid seat and usage-metered billing, per-invoice checkout links with three fee-split modes, idempotent webhook reconciliation.
- Two-way Google and Apple (CalDAV) calendar sync with encrypted OAuth tokens and a retryable row-locked job queue; realtime per-event crew chat on Supabase Realtime.
- Company-enforceable TOTP MFA with hashed recovery codes, tokenized document-access approvals, and CI running ephemeral-Postgres integration tests and Playwright E2E on every PR.

Stack: Next.js, React, Fastify, tRPC, Prisma, PostgreSQL, Supabase, Stripe Connect, Capacitor, Sentry.

Conek, for A-SAFE Digital | Software Engineer I asafe.com | case study **Mar 2025 – Dec 2025**

Internal quote-to-cash platform (quote, sales order, invoice) for A-SAFE, integrated with their CRM, drawing and routing tools.

- Maintained and scaled event-driven microservices on NATS JetStream with type-safe contracts and clean-architecture layering; inter-service latency under 100ms.
- Wrote 500+ unit and integration tests holding 95% coverage, eliminating production regressions.

Stack: TypeScript, Node.js, Fastify, NATS JetStream, Next.js, React, PostgreSQL, AWS.

Sportefy | Trainee Software Engineer App Store | Play Store | case study **Nov 2024 – Feb 2025**

Sports venue booking and matchmaking platform live in Pakistan: Flutter app on both stores, Next.js admin panel, NestJS API.

Flutter app

- 21 feature modules, 823 passing tests, BLoC with clean architecture; offline request queue with replay, automatic token refresh, fault-tolerant parsing.
- QR check-in, wallet checkout, realtime match chat and notifications on Supabase Realtime; server-driven version gate with hard and soft update prompts.

Web admin panel and API

- Next.js admin dashboard over a NestJS API: 48 domain modules, 190+ versioned REST endpoints, additive-only /v1 contract with OpenAPI drift gates in CI; no release broke older installs.
- Payments domain (Paymob behind a pluggable gateway adapter, wallet and credits ledger, promo codes, PDF receipts); multi-actor RBAC from a shared permission matrix; API migrated from EC2 to Railway.

Stack: NestJS, Flutter, Next.js, Prisma, PostgreSQL, Supabase, Paymob, Railway, Vercel, GitHub Actions.

TECHNICAL PROJECTS

RiskSense, Loan-Default MLOps Pipeline **Final-Year Project with Meezan Bank Limited**

- Ensemble of five models orchestrated with ZenML and MLflow (SMOTE balancing, KNN imputation); 92% AUC-ROC with sub-100ms inference from a containerized Flask API.

Stack: Python, ZenML, MLflow, Docker, PostgreSQL, Flask.

Algorithmic Chess Engine

- Minimax with alpha-beta pruning, killer-move heuristics and transposition tables; 75% less search branching, verified 1600 ELO.

Stack: Python.

EDUCATION

Institute of Business Administration (IBA), Karachi **Graduated 2025**
BSc in Computer Science. National Talent Hunt Program Scholar (top 33 of 7,290 candidates).

ACHIEVEMENTS

- **Teaching Assistant, Mobile Application Development (IBA):** mentored 100+ students through hands-on workshops; class assessment averages rose 30%.
- **Kaggle ML challenges:** 2nd and 3rd place across two university-wide competitions (150+ participants).
- **Ignite Hackathon:** 1st place for an IoT software prototype built and deployed within a 2-hour limit.
- **Trading Bazaar:** runner-up for algorithmic trading strategies.